RUDRA SANKAR GHOSH DASTIDAR

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Full-stack developer and published researcher with hands-on experience in React, Next.js, Node.js, Firebase, and WebRTC. Built and deployed cross-platform applications integrating Gemini AI, real-time communication, and machine learning.

Education

VIT-AP University *B.Tech in Computer Science and Engineering* — Amaravati, Andhra Pradesh
CGPA: 9.00 / 10.00 — 2024–2028 (Expected)

Publication

Heart Abnormality Detection System Using Arduino-Based ECG Monitoring — *STAPS Journal • 2025*
Authors: Rudra Sankar Ghosh Dastidar et al.

- Published peer-reviewed research on a portable Arduino-based ECG monitoring system that integrates TensorFlow Lite machine learning for real-time heart abnormality detection from PQRST waveform features.

Projects

HackForge — 2026
React, Node.js, Express.js, MongoDB, JWT, Gemini AI, Tailwind CSS — GitHub | Live

- Built a full-stack hackathon platform supporting project submission, team formation, and multi-user collaboration with role-based access.
- Implemented JWT authentication, RESTful APIs, and MongoDB-backed data management across the application lifecycle.

LifeOnLine — 2026
React Native, Next.js, Node.js, Express.js, Firebase, Gemini AI, WebRTC — GitHub | Live

- Built a cross-platform healthcare application with SOS alerts, location tracking, and WebRTC-powered video consultations.
- Integrated Gemini AI for symptom analysis, AI-driven triage severity scoring, and doctor matching by medical department.
- Implemented Firebase authentication and end-to-end encrypted real-time communication across web and mobile.

NexaBank — 2026
Next.js, TypeScript, JavaScript, CSS — GitHub | Live

- Built a banking web application with authentication, account management, and dashboard using Next.js and TypeScript.
- Architected reusable, type-safe frontend components deployed on Vercel with mobile-first responsive layouts.

Heart Abnormality Detection System — 2025
Arduino UNO, Python, MATLAB, React, Node.js, TensorFlow Lite — GitHub | Live

- Built a portable ECG monitoring device using Arduino UNO, ESP-32, and AD8232 sensors with wireless data transmission.
- Trained a TensorFlow Lite model for binary ECG beat classification; integrated results into a React and Node.js dashboard.

Technical Skills

Languages: Java, Python, C++, JavaScript, TypeScript
Frameworks: React, React Native, Next.js, Node.js, Express.js
Databases: MongoDB, MySQL, Firebase
Tools: Git, GitHub, WebRTC, TensorFlow Lite, Gemini AI, MATLAB, Figma, Vercel

Certifications

- Machine Learning Specialization — DeepLearning.AI (Coursera)
- Entrepreneurship Program — Wadhwani Foundation

Leadership & Activities

Music Lead, BONGOJO — 2025–2026

- Organized cultural events, directed a university drama production, and led the West Bengal contingent at the university State Rally.

Documentation Team Member, Entrepreneurship Club — 2025–2026

- Prepared event reports, proposals, and documentation for club activities.